

SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 06

Exercise 2: Pragma omp (openmp) single code

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On instance with 2 vCPUs (t3.micro)

Nano - creating the pragma omp single code

```
GNU nano 6.2          ex2_single.c
#include <omp.h>
#include <stdio.h>

int main() {

#pragma omp parallel
{
    int th = omp_get_thread_num();
    printf("thread # %d\n", th);

#pragma omp single
    {
        printf("Inside Single # %d\n", omp_get_thread_num());
        printf("Exiting Single \n");
    }
    printf("Hi again from thread # %d\n", omp_get_thread_num());
}

    return 0;
}
```

Outputs from running the pragma omp single code.

```
[ec2-user@ip-172-31-17-201 ~]$ gcc -fopenmp ex2_single.c -o ex2_single.o
[ec2-user@ip-172-31-17-201 ~]$ ./ex2_single.o
thread # 1
Inside Single # 1
Exiting Single
thread # 0
Hi again from thread # 1
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex2_single.o
thread # 1
Inside Single # 1
Exiting Single
thread # 0
Hi again from thread # 1
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex2_single.o
thread # 1
Inside Single # 1
Exiting Single
thread # 0
Hi again from thread # 1
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex2_single.o
thread # 0
Inside Single # 0
Exiting Single
thread # 1
Hi again from thread # 0
Hi again from thread # 1
```

How the `pragma omp single` code works in the program.

`#pragma omp single` ensures that only **one** thread out of the whole batch executes the block inside it and that is whichever thread reaches that point first. Every other thread skips the block entirely.

Unlike `master` (which always specifically assigns the work to thread 0), `single` doesn't care which thread runs it, it is whatever thread reaches it first.

By default, `single` also has an implicit barrier at its closing bracket, meaning all other threads wait there until the one thread executing the block finishes, before any thread proceeds to the "Hi again" line. That's why we never see "Hi again" interleaved with "Inside Single"/"Exiting Single", the barrier keeps them separated.